UNIT 1 · 9 hours

Analysis of the RLC Series Circuit



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

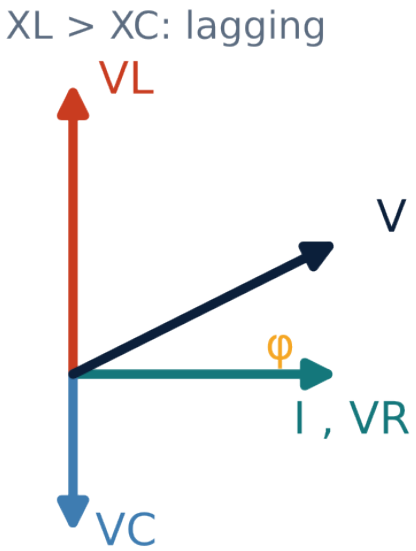
- Draw the phasor diagram for the series RLC circuit
- Find Z, I, φ and voltages across each element
- Classify as inductive, capacitive or resistive
- Explain series resonance

Syllabus context

Analysis of R, L, C, RL, RC and RLC series circuits, real power, reactive power, apparent power and power factor.

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Series RLC phasor diagram



- Reference: current I.
- VR in phase, VL leads by 90°, VC lags by 90°.
- VL and VC are in opposition: net reactive voltage = VL − VC.
- $V = \sqrt{(VR^2 + (VL - VC)^2)} = I Z$

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Impedance of the RLC series circuit

$$X = XL - XC$$

net reactance

$$Z = \sqrt{(R^2 + (XL - XC)^2)}$$

impedance (Ω)

$$\varphi = \tan^{-1}[(XL - XC)/R]$$

phase angle

- XL > XC: net inductive, current lags.
- XL < XC: net capacitive, current leads.
- XL = XC: purely resistive, φ = 0, Z = R (resonance).

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Three cases

Condition	Nature	Current vs voltage	Power factor
XL > XC	Inductive	I lags V	Lagging
XL < XC	Capacitive	I leads V	Leading
XL = XC	Resistive	I in phase	Unity (resonance)

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Numerical: RLC series

Problem R = 10 Ω, L = 0.1 H, C = 100 µF in series across 230 V, 50 Hz. Find Z, I, φ, voltages and P.

- $X_L = 31.42\ \Omega$; $X_C = 31.83\ \Omega$; $X = -0.41\ \Omega$
- $Z = \sqrt{(10^2 + 0.41^2)} = 10.01\ \Omega$
- $I = 230/10.01 = 22.98$ A; $\varphi \approx 2.3°$ leading (almost resonance)
- $V_R = 229.8$ V, $V_L = 722$ V, $V_C = 731$ V
- $P = I^2R = 22.98^2 \times 10$

Answer: I ≈ 23 A, P ≈ 5.28 kW, near-unity pf

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Series resonance

- Condition: XL = XC, so $\omega_0 = 1/\sqrt{(LC)}$ and $f_0 = 1/(2\pi\sqrt{(LC)})$.
- Impedance is minimum (Z = R); current is maximum (I = V/R).
- Voltages VL and VC can far exceed the supply voltage: voltage magnification Q = XL/R.
- Applications: radio tuning, filters.
- Example: L = 0.1 H, C = 100 µF → f_0 = 50.3 Hz.

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Key takeaways

- $X = X_L - X_C$, $Z = \sqrt{(R^2 + X^2)}$
- $X_L > X_C$ lagging; $X_L < X_C$ leading; $X_L = X_C$ resonance
- At resonance $Z = R$, current is maximum, pf = 1
- $f_0 = 1/(2\pi\sqrt{LC})$

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Practice questions

Q1. R = 5 Ω, XL = 12 Ω, XC = 7 Ω. Find Z and φ.

Q2. Find the resonant frequency for L = 20 mH, C = 10 µF.

Q3. A series RLC circuit has R = 20 Ω at resonance on 100 V. Find the current.

Q4. Why can VL exceed the supply voltage?